

M07 Module Review

PostgreSQL + Data Modeling | Review after attempt

Review rule

Open only after a genuine attempt. A review is diagnostic material, not the first way to solve the task.

M07-L01 - Install/verify PostgreSQL and connect safely

Strong evidence: You can name the server version, database and current role from query evidence.

Repair focus: Confusing a GUI connection profile with the database itself or running setup in the wrong database.

M07-L02 - Tables, data types and explicit DDL

Strong evidence: You can explain each column type/constraint from the business contract.

Repair focus: Using text for every field and trusting application code to enforce all validity rules.

M07-L03 - Primary keys, foreign keys and constraints

Strong evidence: You can predict which invalid row should fail and why.

Repair focus: Treating a constraint error as a nuisance and weakening the contract instead of fixing the data/logic.

M07-L04 - Transactions, DML and UPSERT

Strong evidence: You can explain what commits atomically and why a rerun is safe.

Repair focus: Using UPSERT without a stable key or updating every column blindly on conflict.

M07-L05 - Indexes and EXPLAIN evidence

Strong evidence: You can identify the main scan/join operator and explain why the chosen plan is plausible.

Repair focus: Creating indexes on every column or claiming a plan improved just because an index exists.

M07-L06 - Operational vs analytical modeling and normalization

Strong evidence: You can justify where an attribute belongs based on identity, update behavior and query use.

Repair focus: Calling every wide table a star schema or every normalized table unsuitable for analytics.

M07-L07 - Facts, dimensions and star-schema grain

Strong evidence: You can explain every key in the fact and prove aggregation totals reconcile.

Repair focus: Joining a many-valued descriptive table directly to a fact and silently multiplying measures.

M07-L08 - Surrogate keys and referential integrity

Strong evidence: You can trace from fact surrogate key back to the source business key and explain the policy for missing dimensions.

Repair focus: Replacing the source business key entirely and losing the ability to trace back to the source.

M07-L09 - Slowly Changing Dimensions and history

Strong evidence: You can show prior/current versions and explain how a fact chooses the correct version.

Repair focus: Updating the old row in place and destroying history or creating two simultaneous current versions.

M07-L10 - Build and validate a small analytical PostgreSQL database

Strong evidence: All required controls reconcile and can be rerun from version-controlled SQL.

Repair focus: Declaring success because the final chart/query looks plausible without independent reconciliation.

M07-L11 - Bulk loading with COPY and controlled staging

Strong evidence: You can explain the staging-to-target boundary and reconcile file rows to promoted/rejected outcomes.

Repair focus: Bulk loading directly into a final constrained table and then disabling constraints when dirty rows fail.

M07-L12 - Schemas, roles and least-privilege boundaries

Strong evidence: You can state what each role can and cannot do and why.

Repair focus: Using the database superuser for every application/pipeline because permissions are inconvenient.

M07-L13 - Schema change, backup/restore and handoff validation

Strong evidence: Another learner can recreate the schema/data path and verify post-change correctness from saved materials.

Repair focus: Editing production-like schema manually with no migration record or post-change reconciliation.